

CUDDALORE, India

WMO: 433290

Lat: 11.767N

Lon: 79.767E

Elev: 39

StdP: 14.68

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	68.6	69.9	62.5	84.8	76.5	64.1	89.7	76.4	11.0	80.7	9.2	81.2	1.4	340	0.322

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	15.4	100.3	80.5	98.4	80.4	96.7	80.2	84.6	93.0	83.8	92.2	83.2	91.4	4.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.6	170.5	89.4	81.8	165.5	88.8	81.0	161.3	88.2	49.0	93.5	48.1	92.4	47.3	91.8	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
9.6	7.7	6.9	DB	65.2	104.8	1.9	1.9	63.8	106.2	62.7	107.3	61.6	108.4	60.2	109.8	
			WB	61.8	86.1	2.3	1.2	60.2	86.9	58.9	87.6	57.6	88.3	55.9	89.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.6	77.7	79.5	82.8	86.6	89.0	88.8	87.2	86.1	85.2	82.9	79.7	78.0
	DBStd	4.59	1.82	2.04	2.39	2.31	3.03	2.69	2.47	2.32	2.15	2.57	2.17	1.83
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12278	859	826	1015	1098	1210	1164	1155	1120	1055	1018	891	867
	CDD65	6803	394	406	550	648	745	714	690	655	605	553	441	402
	CDH74	80578	3007	3784	6434	8831	10437	9924	9219	8419	7505	6146	3882	2990
	CDH80	37186	680	1173	2657	4595	6050	5671	4882	4173	3443	2320	959	583
Wind	WSAvg	2.5	2.4	2.2	2.3	2.6	3.1	3.1	2.7	2.6	2.3	1.8	2.1	2.6
Precipitation	PrecAvg	54.5	1.0	0.8	0.5	0.3	1.5	1.5	3.6	5.7	5.6	10.6	15.9	7.9
	PrecMax	92.3	8.6	13.7	3.7	2.0	15.0	4.5	8.4	14.4	14.6	24.4	38.9	25.4
	PrecMin	28.9	0.0	0.0	0.0	0.0	0.0	0.2	0.3	0.9	0.1	2.8	4.1	0.2
	PrecStd	14.1	1.9	2.7	0.9	0.4	3.1	1.3	2.2	3.5	3.1	6.0	9.0	6.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.3	90.4	93.7	98.6	103.7	102.2	100.5	98.8	97.0	94.3	89.8	87.1
		MCWB	76.6	77.8	80.1	82.0	81.3	80.3	79.5	79.6	78.6	79.3	78.8	77.2
	2%	DB	86.2	88.6	91.9	95.4	100.6	99.9	98.2	96.5	94.9	92.0	87.8	85.6
		MCWB	75.4	77.1	80.0	82.4	81.3	80.1	79.3	79.4	79.4	79.6	78.4	76.3
	5%	DB	85.1	87.2	90.7	93.9	98.2	98.0	96.4	94.7	93.3	90.5	86.4	84.5
		MCWB	74.8	76.3	79.5	82.2	81.5	80.1	79.2	79.1	79.5	79.6	77.8	75.3
	10%	DB	83.9	86.0	89.4	92.4	96.0	96.1	94.5	92.9	91.5	88.8	85.1	83.4
		MCWB	74.2	75.6	78.8	81.7	81.8	80.0	79.1	79.3	79.8	79.4	77.5	74.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.9	80.0	82.7	85.2	86.1	85.1	84.2	83.5	82.8	82.5	81.2	79.8
		MCDB	85.5	87.7	89.9	92.5	93.9	94.5	93.4	92.6	90.9	88.8	86.3	84.2
	2%	WB	77.4	78.8	81.7	84.1	84.8	83.8	82.6	82.4	81.9	81.6	80.2	78.7
		MCDB	83.2	86.0	88.7	92.0	93.0	93.4	91.5	90.2	89.3	87.9	85.2	82.9
	5%	WB	76.5	77.9	80.9	83.3	84.1	82.8	81.6	81.6	81.4	80.9	79.5	77.7
		MCDB	82.0	84.7	87.9	90.9	92.5	92.2	90.4	89.5	88.7	87.3	84.1	81.8
	10%	WB	75.6	77.1	80.1	82.5	83.4	81.9	80.8	80.9	80.8	80.2	78.8	76.8
		MCDB	81.1	83.4	87.0	89.9	91.6	91.2	89.5	88.8	88.0	86.4	83.1	80.7
Mean Daily Temperature Range	5% DB	MDBR	13.3	14.5	14.3	13.1	15.4	16.1	15.5	14.9	13.9	11.7	10.2	11.1
		MCDBR	14.2	14.9	14.7	14.3	18.1	18.2	17.9	17.1	16.1	14.1	12.6	12.8
		MCWBR	5.8	5.7	5.3	5.2	6.8	6.3	6.5	6.4	6.0	5.3	5.3	5.5
	5% WB	MCDBR	12.1	13.1	13.4	13.6	15.7	16.8	16.3	15.4	14.3	12.5	10.3	9.9
		MCWBR	5.7	5.5	5.3	5.5	6.6	7.3	7.3	6.9	6.2	5.6	5.2	5.3
Clear-Sky Solar Irradiance	taub	0.460	0.453	0.495	0.515	0.594	0.525	0.532	0.522	0.516	0.538	0.508	0.503	
	taud	2.207	2.228	2.114	2.109	1.908	2.122	2.099	2.132	2.149	2.074	2.133	2.105	
	Ebn at Noon	267	274	265	257	233	246	246	252	255	247	251	251	
	Edh at Noon	45	46	52	52	62	50	51	50	50	53	48	49	
All-Sky Solar Radiation	RadAvg	1588	1875	2027	2033	1926	1783	1695	1728	1735	1497	1310	1330	
	RadStd	66	71	91	88	123	120	111	89	109	89	142	117	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+0.50	N/A	N/A	+1.12	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page